Zeynep Çaysar



MSc student majoring in Information Systems with a Data Science minor at the University of Zurich.
Interested in data analytics, data science, ML, AI, LLMs, information systems and software development.

Email: zeynepcaysar@gmail.com
GitHub: GitHub **LinkedIn:** zeynepcaysar
Personal Website: zeynepcaysar.github.io
Languages: Turkish (Native), English (C1), French (C1)

Relevant skills

Programming languages: Python, Scala, JavaScript

Python: Pandas, Numpy, Seaborn, Matplotlib, Scikitlearn, Scipy

Python for Backend Development: Flask, FastAPI, API design & integration, JSON

Machine Learning models: Logistic Regression, Linear Regression, Decision Trees, Random Forest, Gradient Boosting, KNN, DNN, LLMs

Database Management: SQL

Web Development: HTML, CSS, React JS

Version Control and Collaboration: GitLab, GitHub, Jira, Kanbanize

Documentation & Reporting Tools: LaTeX, Marp, Microsoft Office (Excel, Word, PowerPoint)

Development Environments: Linux, VS Code, PyCharm, Jupyter Notebooks

Experience

Openlab Intern, CERN, Geneva, Switzerland
July 2025 - August 2025

- Developed a plugin prototype for Indico, CERN's open-source event management platform, to automatically summarize meeting minutes using Hugging Face LLMs.
- Built a frontend component in React (with Indico's UI framework) to allow users to trigger and customize summarization.
- Tested multiple open-source models (e.g. LLaMA-3, Mistral, Gemma, Phi, Qwen) for accuracy, latency, and hallucination.
- Collaborated with the Indico development team.
- Delivered a presentation and wrote a report.
- **Technologies:** Python, Flask, HTML, CSS, JavaScript, React JS, Hugging Face Transformers, GitHub
- **Project Repository:** GitHub

Education

Master of Science in Information Systems, minor in Data Science, University of Zurich

Feb 2025 - Jun 2027
Research Project: Computing Similarities Between Patient and Rare Disease Profiles
Feb 2026 - Sep 2026

- Developed a framework for ranking rare diseases based on patient phenotypes using ontology-based, graph-based and machine learning similarity methods.
- Implemented and evaluated multiple approaches, including semantic similarity, TF-IDF, transformer embeddings, HPO2Vec, autoencoders, and LLMs.
- Integrated biomedical ontologies and disease databases (HPO, MONDO, ORDO) to construct standardized disease profiles and support phenotype-based retrieval.
- Benchmarked methods across diverse patient datasets using Recall@K, MRR, and ensemble ranking techniques (Reciprocal Rank Fusion).
- Developed a web application using FastAPI and Vue.js to allow users to input patient phenotypes and retrieve ranked rare disease candidates.
- **Technologies:** Python, Scikit-learn, PyTorch, Hugging Face Transformers, FastAPI, JavaScript, Vue.js, Git
- **Project Repository:** GitHub

Bachelor of Science in Information Systems and Service Science, University of Geneva

Sep 2021 - Jan 2025
Final Grade: 5.05/6
Relevant Courses: Introduction to programming, Formal languages, Modeling information systems and services, Computer operating principles, Applied algorithms, Databases, Advanced databases, Software and computer networks, Communication networks, Project management, Human-computer interface, Multimedia services and technologies, Security of information systems, Probabilities, Introduction to statistics
Bachelor's Thesis: Exploratory Data Analysis and Predictive Modeling of Migraine Attacks and Quality of Life: Using Physical Activity, Sleep, and Stress Patterns
Thesis Grade: 6/6

- Analyzed large-scale migraine datasets (2 GB, 69+ variables) to investigate relationships between sleep, physical activity, stress, and migraine outcomes using statistical analysis and visualization.
- Developed and evaluated machine learning models, including Random Forest, Gradient Boosting, Logistic Regression, and neural networks, to predict migraine-related outcomes.
- Built an interactive web application using Flask to predict migraine pain intensity based on patient inputs.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, TensorFlow, SciPy, Flask, HTML/CSS, GitLab
- **Project Repository:** GitHub

Conferences & Research presentations

Towards Digital Biomarkers of Migraine: Preliminary Findings from the COMPUTEMIGRAINE Study
International Headache Congress (IHC 2025) · Abstract IHC25-DP-089
Researchers: Anna Ferrari, Zeynep Çaysar, Katarzyna Wac

- Presented the research abstract as both a digital poster and a printed poster.